Replication materials are great but software versioning still poses a problem for open science

Taylor J. Wright* Bruno Rodrigues †

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*Brock University, twright3@brocku.ca
†Ministry of Research and Higher education, Luxembourg

1 Introduction

Beginning in the early 2010s with psychology and quickly spilling over into political science and economics on the backs of high-profile cases of fraud (Broockman, Kalla, and Aronow 2015) and errors (Herndon, Ash, and Pollin 2014), the "replication crisis" in the social sciences has become mainstream, even finding its way into the popular press (Aschwanden 2015; Gelman 2018). Concerns over how well published results hold up to replication in other settings has led to large scale initiatives to document how trustworthy the existing stock of evidence is (e.g. Camerer et al. 2016; Chang and Li 2022; and Open Science Collaboration 2015). While the *replicability* of studies has received much attention, there has been a tandem movement discussing the importance of and need for research *reproducibility*. While the exact scope and definition is subject to ongoing debate, replication is, broadly speaking, re-testing a hypothesis while changing an element of previous research (e.g. the sample or estimating equation) whereas reproducibility (sometimes referred to as verification or computational reproducibility) is following a study's protocol or methods exactly and obtaining the results presented in the study.[1]

In our view, reproducibility is an insufficient but necessary condition for replication—that is, work cannot be replicable if it is not reproducible and it likely does not make sense to spend resources on replication if research is not in the first place reproducible.

These concerns about the reproducibility (and replicability) of social science research have prompted a push, often referred to as Data Access and Research Transparency (DA-RT) in political science following Lupia and Elman (2014), for journals to require the publication of research materials that accompany academic research.[2] Specifically, the provision and publication of underlying data and code used for data preparation and analysis. This push has seen some success, with a growing number journals now requiring the provision of replication materials as a condition of publication. Some journals, such as the American Economic Review (AER) or American Journal of Political Science (AJPS), even have verification policies that require authors to upload their replication materials and have their results verified by another team (either the journal's replication team (AER) or a third party (AJPS)) before publication. However, even if these materials are provided, and even when in place

[1] For further discussion on definitions, scopes, and types of replication and reproducibility see, for example, Clemens (2017), Derksen and Morawski (2022), Hamermesh (2007), and Nosek and Errington (2020).

[2] See, for example, the 2014 Joint Editors Transparency Statement which was signed by editors of 27 leading political science journals: https://www.dartstatement.org/2014-journal-editors-statement-jets

these policies do not have perfect compliance (Philip 2010; Stockemer, Koehler, and Lentz 2018), regular software updates and new version releases can result in the replication materials failing to faithfully reproduce the authors' results or even run at all. [3]

In this paper, we present a case study of an article published in Journal of Politics in January 2022 titled, "Multiracial Identity and Political Preferences", (Davenport, Franco, and Iyengar 2022), that details replication challenges arising from changes in the statistical software R. We were unable to reproduce the authors' results using either the current version of R, or the version that the authors indicate they used. The lack of reproducibility arose due to a change in the defaults used by base R when generating random numbers starting in version 3.6.0.

We contribute to the existing literature in three ways. First, we add to the discussion about the importance of the availability of replication materials. Existing research documents that the availability of replication materials is associated with a higher citation count (Piwowar, Day, and Fridsma 2007; Piwowar and Vision 2013) and with a higher likelihood of replication (Philip 2010). Much of this research focuses on the provision of data and code as a necessary condition for facilitating replication and rebuilding trust in science (Dafoe 2014; Lupia and Alter 2014). We contribute to this literature by emphasizing that, while we agree that data and code availability are necessary, their existence cannot guarantee reproducibility. This brings us to our second contribution: we expand on the discussion of problems raised by software and package versioning for reproducibility by providing a walkthrough of the problem using a concrete example. Lastly, we further the discussion of tools and best practices for ensuring reproducibility. Specifically, we provide a step-by-step guide to using Docker and the `renv` R package to ensure the reproducibility of research.

The rest of the article proceeds as follows: Section 2 walks through the reproducibility issues in Davenport, Franco, and Iyengar (2022); Section 3 discusses currently available tools and best practices (e.g. Docker and R packages such as `renv`, `groundhog`) for ensuring that replication materials continue to faithfully reproduce research results, despite post-publication changes in the tools used; and Section 4 concludes.

[3] Simonsohn (2021) presents several examples of R changes that could break scripts.

2 Reproduction

2.1 Illustrating the issue with software versioning

Davenport, Franco, and Iyengar (2022) examine the political ideology and policy preferences of multiracial Americans (those who identify with more than one racial group). This section of Americans is growing rapidly, but little is known about their politics. The authors survey White-Asians and White-Blacks from YouGov's Internet panel and compare them to similarly surveyed White, Black, and Asian monoracial respondents, looking for differences in political ideology; party affiliation;

positions about cultural issues, social welfare, and police treatment across racial groups.

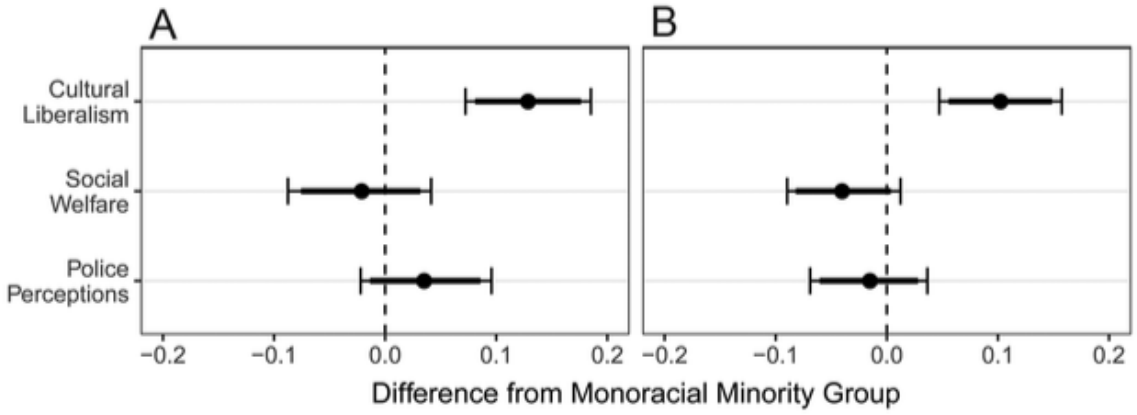

Figure 1: Figure 1 from Davenport, Franco, and Iyengar (2022). Multiracial attitudes relative to monoracial minorities. A, White-Asian; B, White-Black. Higher values are more liberal attitudes.

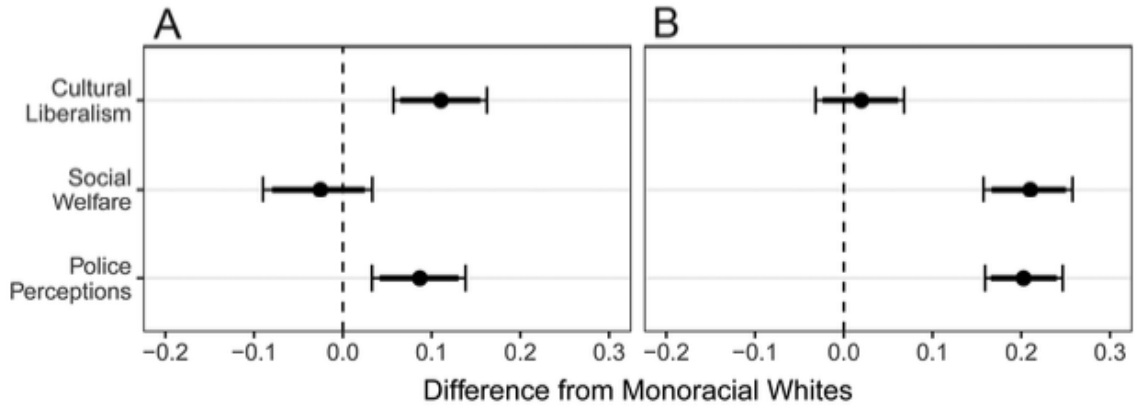

Figure 2: Figure 2 from Davenport, Franco, and Iyengar (2022). Multiracial attitudes relative to monoracial Whites. A, White-Asian; B, White-Black. Higher values are more liberal attitudes.

Figures 1 and 2 present Figures 1 and 2 from Davenport, Franco, and Iyengar (2022) which contain the main results. Panel A corresponds to results for White-Asian respondents while panel B corresponds to White-Black respondents. In Figure 1, the x-axis reflects the differences in a given multiracial group's attitudes relative to the corresponding monoracial group (e.g. White-Asian relative to Asians) with higher values meaning more liberal attitudes. In Figure 2, the x-axis reflects a similar difference but relative to monoracial Whites (e.g. White-Asian relative to Whites). Figure 1 indicates that White-Asians have more liberal attitudes on cultural issues than Asians while White-Blacks have more liberal attitudes on cultural issues than Blacks but views regarding social welfare and police perception appear to be comparable between White-Asians and Asians and between White-Blacks and Blacks. Figure 2 Panel A suggests that White-Asians are more liberal than Whites when it comes to cultural issues and police perceptions but similarly liberal with views on social welfare. Figure 2 Panel B indicates that White-Blacks are similarly liberal to Whites on cultural issues but are more liberal on issues of social welfare and police perceptions.

Importantly, the authors provide replication materials at the Journal of Politics Dataverse, including the code used to conduct their analysis as well as the data itself. The code files contain comments noting that the analysis "Requires R version 3.6.2 (2019-12-12)".

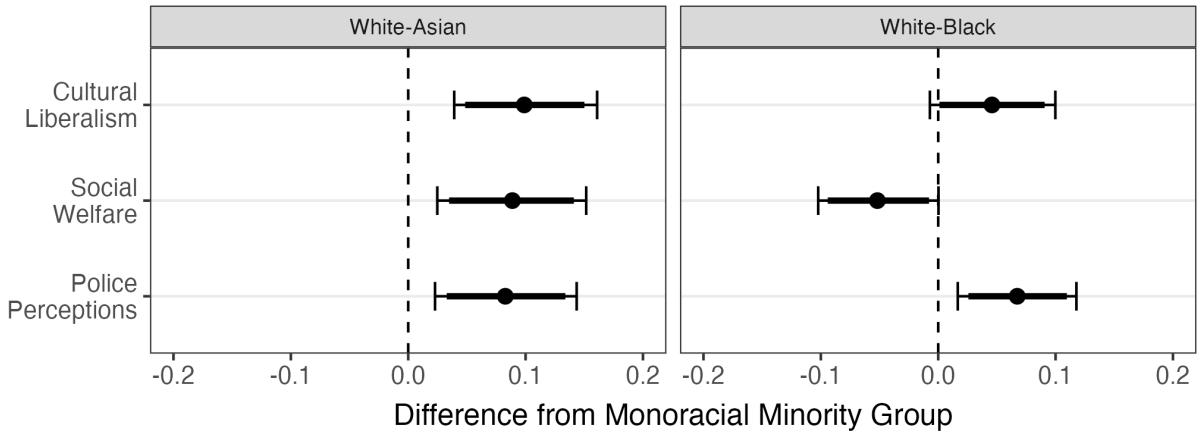

Figure 3: Figure 1 from Davenport, Franco, and Iyengar (2022) when using a later software version (R post 3.6.0). Multiracial attitudes relative to monoracial minorities. A, White-Asian; B, White-Black. Higher values are more liberal attitudes.

Figures 3 and 4 present Figures 1 and 2 from Davenport, Franco, and Iyengar (2022) when using a later software version (specifically a version of R post 3.6.0 such as the version 3.6.2 that the authors note in their code files). In Panel A of Figures 3 and 4, White-Asians are now more liberal than Asians and Whites on all issues. In Panel

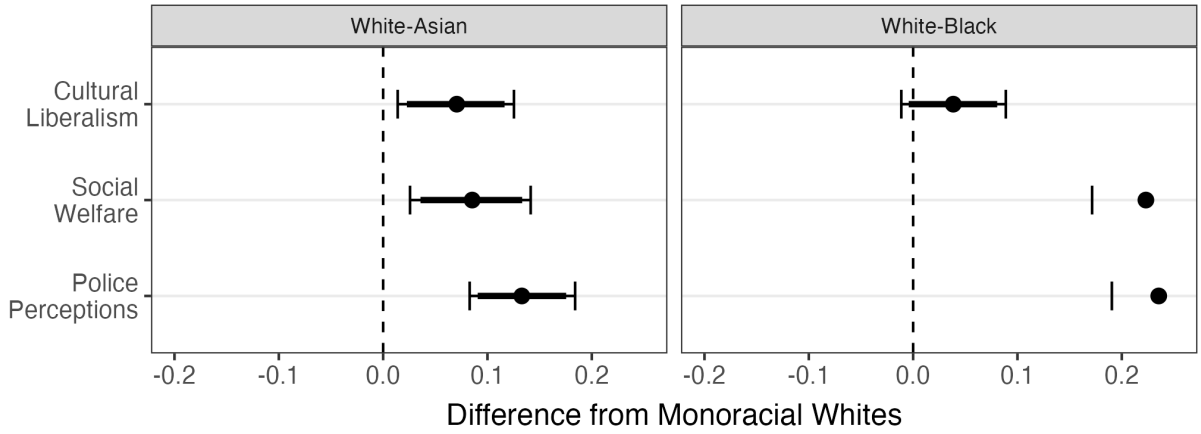

Figure 4: Figure 2 from Davenport, Franco, and Iyengar (2022) when using a later software version (R post 3.6.0). Multiracial attitudes relative to monoracial Whites. A, White-Asian; B, White-Black. Higher values are more liberal attitudes.

B of figure 3 White-Blacks have more liberal views on cultural issues than Blacks (in terms of point estimates though the confidence interval now contains 0) and are now more liberal regarding police perceptions. In Panel B of figure 4 the point estimates appear very similar though there is an issue computing confidence intervals.

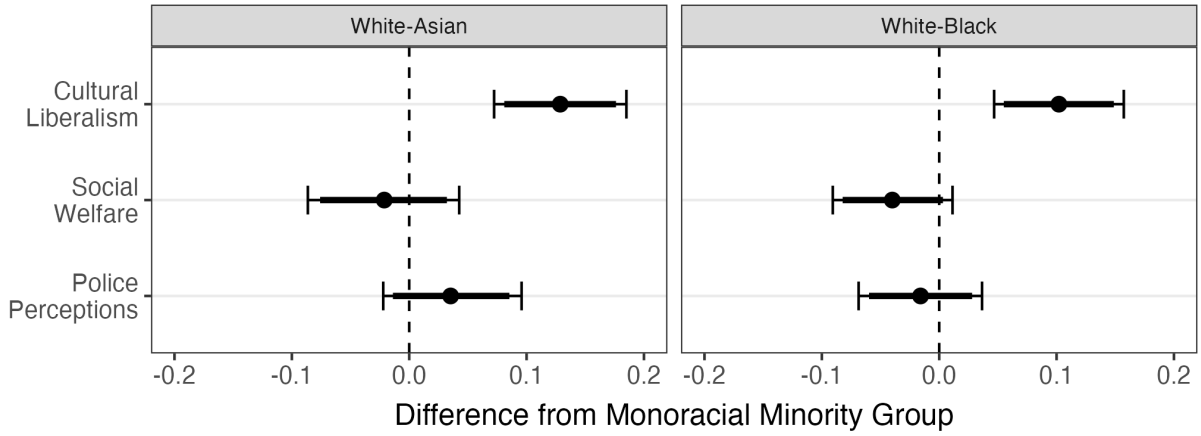

Figure 5: Figure 1 from Davenport, Franco, and Iyengar (2022) when using an earlier software version (R pre 3.6.0). Multiracial attitudes relative to monoracial minorities. A, White-Asian; B, White-Black. Higher values are more liberal attitudes.

Figures 3 and 4 provide Figures 1 and 2 from Davenport, Franco, and Iyengar (2022) when using an earlier software version (specifically a version of R pre 3.6.0). Minor issues with the confidence intervals in Panel B of Figure 6 aside, these are otherwise the same as those presented in the published article.

The issue seems to be that the weights the authors use are non-integer and the main function used in their analy-

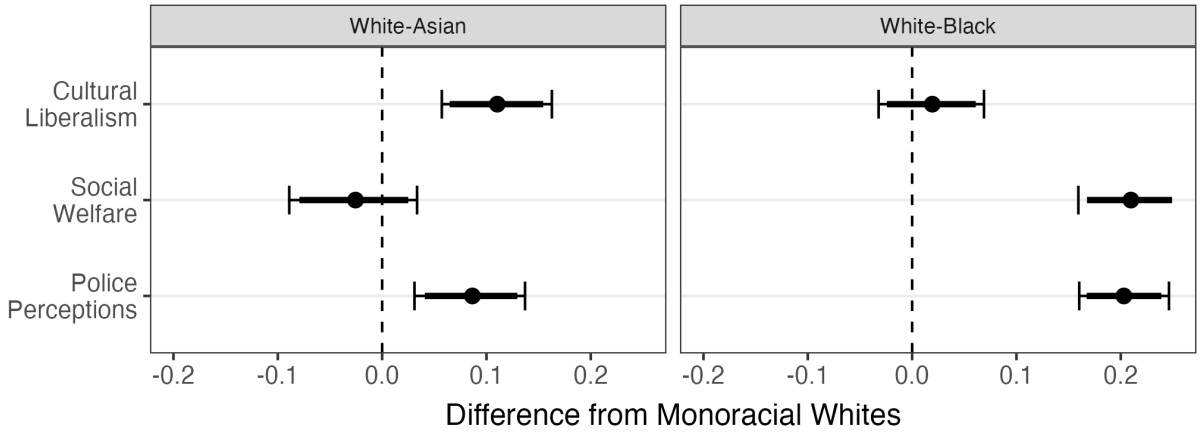

Figure 6: Figure 2 from Davenport, Franco, and Iyengar (2022) when using an earlier software version (R pre 3.6.0). Multiracial attitudes relative to monoracial Whites. A, White-Asian; B, White-Black. Higher values are more liberal attitudes.

sis, `Zelig()`, uses `sample()` in cases with non-integer values (see http://docs.zeligproject.org/articles/weights.html). Following changes to base `R` in 3.6.x the sample() function, and thus now Zelig, yields different results in `R` versions later than 3.5.x. Before R 3.6.0 `RNGkind(sample.kind = "Rounding")` was the default behaviour but after 3.6.0 the sample function's new default behaviour is `RNGkind(sample.kind = "Rejection")` (see https://blog.revolutionanalytics.com/2019/05/whats-new-in-r-360.html for more details).

The key takeaway is that despite the authors providing replication materials and indicating which version of `R` they believed they used, the results cannot be reproduced. This is not a criticism of the authors, but rather a demonstration of the challenges of reproducibility in the face of software updates. There are ways to ensure that the estimates are consistent (such as ensuring to always explicitly articulate options in functions so that changes to defaults cannot break existing code. However, reproducibility is still threatened by software updates that result in larger changes (such as what inputs are required). Manually documenting software versions (including packages or commands) in the code or README files is a good start, but as our example shows, it is not enough to ensure faithful reproduction.

3 Discussion

3.1 Rebuilding the original development environment using Docker

Replicating results from past studies is quite challenging, for many reasons. This article focuses on one of these reasons: results cannot be reproduced because of changes introduced in more recent versions of the software used for analysis, despite the availability of both data and the author' original scripts.

To reproduce the results from the original study and to pinpoint the impact of the change introduced in R 3.6.0, we chose to use Docker. Docker is a containerization tool that enables one to build so-called *images*. These images are usually built on top of a Linux operating system and contain all of the required software to build any type of output. After the image containing the output has been built, users in turn only need to be able to run Docker *containers* instantiated from the image definition to build the output. Containerization tools such as Docker solved the "works on my machine" problem: this problem arises when software that works well on the development machine of the developer fails to run successfully on a customer's machine. This usually happens because the customer's computer does not have the necessary dependencies to run the software (which are traditionally not shipped alongside the software product) or because of a version mismatch of the operating system.

The same approach can be used to build research projects, which also suffer from the same "works on my machine" problem as any other type of software. Containerization tools offer a great opportunity for reproducibility in research: instead of just sharing a replication script, authors can now easily share the right version of the software used to produce these scripts, as well as the right version of the used libraries by building and sharing a Docker image (or at least provide the necessary blueprint to enable others to do so, as we will discuss below). Future researchers looking to reproduce the results can now simply run a container from the provided image (or build an image themselves if the original authors provided the required blueprints).

Concretely, to build a Docker image, a researcher writes a so-called *Dockerfile*, which lists all the necessary steps to build an output. Here is an example of a very simple Dockerfile:

```
FROM rocker/r-ver:4.3.0
```

```
CMD ["R"]
```

This Dockerfile contains two lines: the first line states which Docker image we are going to use as a base. Our image will be based on the `rocker/r-ver:4.3.0` image. The Rocker project (Boettiger and Eddelbuettel 2017) is a repository containing many images that ship different versions of R and packages pre-installed: so the image called `r-ver:4.3.0` is an image that ships R version 4.3.0. The last line states which command should be run when the user runs a container defined from our image, so in this case, simply the R interactive prompt. Concretely, the above image would allow someone to run R version 4.3.0 as a container, completely isolated from the rest of their computer. This allows users to run different versions of R on the same system. We can add more commands to run a complete research project, from start to finish. Below is an example of a Dockerfile that runs an analysis script, by first stating which base image is going to be used, and then adding packages, files and data to run the entire project:

```
FROM rocker/r-ver:4.3.0

RUN R -e "install.packages('dplyr')"

RUN mkdir /home/research_project

RUN mkdir /home/research_project/data

RUN mkdir /home/research_project/project_output

RUN mkdir /home/research_project/shared_folder

COPY analyse_data.R /home/research_project/analyse_data.R

COPY dataset.csv /home/research_project/dataset.csv
```

```
RUN cd /home/research_project && R -e "source('analyse_data.R')"

CMD mv /home/research_project/project_output/* /home/research_project/shared_folder/
```

This Dockerfile starts from the same base image, an image that ships R version 4.3.0, then it installs the `{dplyr}` package, a popular R package for data manipulation and it creates three directories:

- `/home/research_project`
- `/home/research_project/project_output`
- `/home/research_project/shared_folder`.

Then, it copies the `analyse_data.R` script, which contains the actual analysis made for the research project, and the `dataset.csv` file, into the Docker image. The second-to-last line runs the script, and the last line moves the outputs generated from running the `analyse_data.R` script to a folder called `shared_folder`. It is important to say that `RUN` statements will be executed as the image gets built, and `CMD` statements will be executed as a container runs. Using this Dockerfile, an image called `research_project` can be built with the following command:

```
docker build -t research_project .
```

This image can then be archived and shared for replication purposes. Other images can be built on top of this one, and will thus contain R version 4.3.0, the `{dplyr}` package, the data and the project's folder structure. Future researchers can also run a container from that image using a command such as:

```
docker run -d -it --rm --name research_project_container \
  -v /host/machine/shared_folder:/home/research_project/shared_folder:rw \
  research_project
```

The container, called `research_project_container` will execute the `CMD` statement from the Dockerfile, in other words, move the outputs to the `shared_folder`. This folder is like a tunnel between the machine that

runs the container and the container itself: by doing this, the outputs generated within the container can now be accessed from the host's computer.

The image built by the process above is immutable: so as long as users can run it, the outputs produced will be the same as when the original author ran the original analysis. However, if the image gets lost, and needs to be rebuilt, the above `Dockerfile` will not generate the same image. This is because the `Dockerfile`, as it is written above, will download the version of `{dplyr}` that is current at the time it gets built. So if a user instead builds the image in 5 years, the version of `{dplyr}` that will get downloaded will not be the same as the one that was used for the original analysis. The version of `R`, however, will forever remain at version 4.3.0.

So to ensure that future researchers will download the right versions of packages that were originally used for the project, the original researcher also needs to provide a list of packages that were used as well as the packages' versions. This can be quite tedious if done by hand, but thankfully, there are ways to generate such lists very easily. The `{renv}` package for the `R` programming language provides such a function. Once the project is done, one simply needs to call:

```
RUN rig install 4.2.0

RUN rig default 4.2.0
```

Everything else: package versions and operating system that the replication script runs on, stayed the same.

With this setup, we were able to run the original analysis on an environment that was as close as possible to the original environment used by Davenport, Franco, and Iyengar (2022). However, it is impossible to regenerate the exact environment now. As stated, we had to make an assumption on the date the packages were downloaded, but the packages used by the original authors might have been much older. Another likely difference is that the operating system used inside Docker is the Ubuntu Linux distribution. While Ubuntu is a popular Linux distribution, it is much more likely that the original authors used either Windows or macOS to perform their analysis. In most cases, the operating system does not have an impact on results, but there have been replication studies that failed because of this, as in Bhandari Neupane et al. (2019). Thankfully, the mismatch of the operating system does not seem to be an issue here.

It should also be noted that strictly speaking, using Docker is not completely reproducible either: this is because the underlying base image, upon which we build our complete analysis, also changes through time. In the examples above the image we used, `rhub/rig:latest` is based on Ubuntu 22.04, which is, as of writing, the latest long-term support release of the Ubuntu operating system. This version of Ubuntu thus still gets updated: so when building the image for our project, the underlying base image will be different today than 6 months from now. And when the next Ubuntu long-term support will be released, in April 2024, then `rhub/rig:latest` will this time be based upon this new release of Ubuntu. So building in May 2024 will yield a different image than now. To avoid this, it is possible to use a so-called *digest* which ensures the same image will forever be used as a base. For this paper, we used the following image: `rhub/rig@sha256:0667321839be0c2ade5c14fd8c8bbb311a843ddf690081c5d0dd80dcc98bd6c6`.

Finally, to ensure completeness, instead of using the Posit Package Manager with a fixed date, we could have used the `{groundhog}` package by Simonsohn (2023). This package makes it possible to download packages as of a certain date as well, and it does not require users to change any of `R`'s default configurations.

[4] https://github.com/r-lib/rig
[5] https://packagemanager.posit.co/client/#/repos/2/overview

4 Conclusion

Researchers have to deal with the following aspects of their projects to ensure their long-term reproducibility:

- They need to state the correct versions of R (or any other language) used for the analysis;
- They need to state the correct versions of every other used package for the analysis;
- They need to make sure that their code is readable and easy to explore; and
- They need to provide instructions on how to run the whole project.

By using Docker, it is possible to provide the correct versions of `R` and packages (using `renv`, `groundhog` or a snapshot of the CRAN repository). Another important way to improve reproducibility is also to set up research projects as the output of pipelines instead of scripts, but this is outside the scope of this paper. But this is not only useful for making projects reproducible: by using these tools, researchers also avoid common pitfalls while working on their projects. Research projects can last for years and researchers typically work on several projects in parallel; by using the tools discussed in this article, researchers can thus have project-specific `R` and package versions, thus avoiding the common issue of updating their package library because they need the latest feature of a package for one project, but because packages got updated, another project now produces different results or simply does not run anymore.

However, we realize that the entry cost to a tool like Docker is not low: so at the very least, we advise researchers interested in reproducibility to be very conservative with updating R itself and to use either `renv` or `groundhog` to have project-specific libraries. This is because if researchers at least provide the R version used and an renv.lock file (if they used `renv`, or a date, if they used `groundhog`) it is always possible to re-generate an environment with the right version of `R` and packages as we did for this paper. This solution is not the gold standard of reproducibility though, but easy enough to be implemented at a low cost. This also implies that researchers need to share their code and all other necessary information on a platform such as Github, and it implies that research has to be performed using free and open source tools as proprietary tools cannot easily be installed by other researchers wanting to reproduce a study.

A final issue that we want to address is the over-reliance on Docker. While we have several alternatives for handling packages, Docker was the only solution that we suggested to essentially "freeze" versions of R and other lower-level system dependencies. We want to finish this conclusion by stating that other tools achieve the same purpose, notably Nix (Dolstra 2006). Nix is not a containerization tool like Docker, but a package manager that allows users to install exact versions of software, including scientific software, reproducibly. It is possible to essentially use Nix to replace Docker and `renv` (or `groundhog`), but presenting this tool is outside the scope of this article.

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